

I had a couple of slip-ups the first attempt that I was able to resolve here, overall a very satisfying sample of questions that didn't pose too much trouble. I used my calculator for fraction arithmetic so that I could carry angles through the solution before collapsing to decimals at the final step. I also used my calculator for the trigonometric ratios.

7/29/25, 5:29 PM

SIT292 Module 3 Self-Assessment

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Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	Review
Question 1	4/4	
Question 2	1/1	
Question 3	1/1	
Question 4	1/1	
Question 5	1/1	
Question 6	1.5/2	
Total	9.5/10	(95%)

Congratulations! You have achieved the minimum threshold for this module's self-assessment.

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

Performance Summary

Exam Name:	SIT292 Module 3 Self-Assessment
Session ID:	2350677760761549
Exam Start:	Tue Jul 29 2025 16:25:20
Exam Stop:	Tue Jul 29 2025 17:29:29
Time Spent:	1:04:09

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<https://www.newcastle.ac.uk>).

Question 1

Starting with the matrix,

$$A = \begin{pmatrix} 2 & 2 & 3 \\ 4 & 3 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

The results following the first steps of the inversion algorithm are as follows:

Augmented matrix

$$\begin{pmatrix} 2 & 2 & 3 & 1 & 0 & 0 \\ 4 & 3 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{pmatrix}$$

Step 1.

$$\begin{pmatrix} 1 & 1 & 2 & 0 & 0 & 1 \\ 4 & 3 & 1 & 0 & 1 & 0 \\ 2 & 2 & 3 & 1 & 0 & 0 \end{pmatrix}$$

Step 2.

$$\begin{pmatrix} 1 & 1 & 2 & 0 & 0 & 1 \\ 0 & -1 & -7 & 0 & 1 & -4 \\ 0 & 0 & -1 & 1 & 0 & -2 \end{pmatrix}$$

Write down the resulting augmented matrix after performing the next step (step 3).

Rows: Columns:

$$\begin{pmatrix} 1 & 0 & -5 & 0 & 1 & -3 \\ 0 & 1 & 7 & 0 & -1 & 4 \\ 0 & 0 & -1 & 1 & 0 & -2 \end{pmatrix} \checkmark$$

Expected answer:

$$\begin{pmatrix} 1 & 0 & -5 & 0 & 1 & -3 \\ 0 & 1 & 7 & 0 & -1 & 4 \\ 0 & 0 & -1 & 1 & 0 & -2 \end{pmatrix}$$

(Note that the step by step version of this problem is available in sub-module 3.2. At Step 3, you should not perform any additional operations to obtain a leading 1 in row 3 -- that would be part of Step 4)

Based on this and the Step 4 that would follow, the inverse matrix is:

$$A^{-1} =$$

Rows: Columns:

$$\begin{pmatrix} -5 & 1 & 7 \\ 7 & -1 & -10 \\ -1 & 0 & 2 \end{pmatrix} \quad \checkmark$$

Expected answer: $\begin{pmatrix} -5 & 1 & 7 \\ 7 & -1 & -10 \\ -1 & 0 & 2 \end{pmatrix}$

Score: 4/4 

Gap 0

 Your answer is correct. You were awarded **2** marks.

Gap 1

 Your answer is correct. You were awarded **2** marks.

You scored **4** marks for this part.

Question 2

For the following elementary matrix, indicate the corresponding elementary row operation and write the inverse.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{pmatrix}$$

Row operation applied

- ☐ I. Interchange two rows
- ☐ II. Multiply one row by a nonzero number
- ☒ III. Add a multiple of one row to a different row



Expected answer:

- ☐ I. Interchange two rows ☐ II. Multiply one row by a nonzero number
- ☒ III. Add a multiple of one row to a different row

Inverse matrix

Rows: Columns:

1	0	0
0	1	6
0	0	1



Expected answer:

1	0	0
0	1	6
0	0	1

Score: 1/1 ✓

Gap 0

✓ You chose a correct answer. You were awarded **0.5** marks.

Gap 1

✓ Your answer is correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

Question 3

For the following elementary matrix, indicate the corresponding elementary row operation and write the inverse.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{7} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Row operation applied

- ☐ I. Interchange two rows
- ☒ II. Multiply one row by a nonzero number
- ☐ III. Add a multiple of one row to a different row



Expected answer:

- ☐ I. Interchange two rows
- ☒ II. Multiply one row by a nonzero number
- ☐ III. Add a multiple of one row to a different row

Inverse matrix

Rows: Columns:

1	0	0
0	7	0
0	0	1



Expected answer: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Score: 1/1 ✓

Gap 0

✓ You chose a correct answer. You were awarded **0.5** marks.

Gap 1

✓ Your answer is correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

Question 4

Find a matrix, E , such that $B = EA$ if

$$A = \begin{pmatrix} -5 & 9 \\ -7 & -6 \end{pmatrix}, B = \begin{pmatrix} -26 & -9 \\ -7 & -6 \end{pmatrix}$$

$E =$ Rows: Columns:

$$\begin{pmatrix} \boxed{1} & \boxed{3} \\ \boxed{0} & \boxed{1} \end{pmatrix}$$



Expected answer: $\begin{pmatrix} \frac{1}{0} & \frac{3}{1} \end{pmatrix}$

Score: 1/1 ✓

✓ Your answer is correct. You were awarded 1 mark.

You scored 1 mark for this part.

Question 5

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation.

Find $T \begin{pmatrix} -2 \\ -8 \\ -6 \end{pmatrix}$ if $T \begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 10 \\ -20 \end{pmatrix}$ and $T \begin{pmatrix} -1 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ -14 \end{pmatrix}$.

Enter your result below.

$$T \begin{pmatrix} -2 \\ -8 \\ -6 \end{pmatrix} =$$

Rows: Columns:

$$\begin{pmatrix} \boxed{-14} \\ \boxed{12} \end{pmatrix}$$



Expected answer: $\begin{pmatrix} -14 \\ 12 \end{pmatrix}$

Score: 1/1 ✓

✓ Your answer is correct. You were awarded 1 mark.

You scored 1 mark for this part.

Question 6

Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and the effect of multiplying $T \begin{pmatrix} x \\ y \end{pmatrix}$.

a)

If

$$T = \begin{pmatrix} -\frac{4}{5} & -\frac{3}{5} \\ \frac{-6}{10} & \frac{4}{5} \end{pmatrix}$$

This linear mapping is

- ☐ a rotation about the origin by θ
- ☒ a reflection through the line $y = mx$



Expected answer:

- ☐ a rotation about the origin by θ
- ☒ a reflection through the line $y = mx$

Give the value of θ if it is a rotation or the value of m if it is a reflection

-3 ✓

Expected answer: -3 -3

*Give values of θ in terms of 'pi', e.g., $\pi/2$, $3\pi/5$ etc and express over the domain $(-\pi, \pi]$.

**If your value involves surds, use 'sqrt', e.g. sqrt(2) for the square root of 2.

Score: 1/1 ✓

▼ Hide feedback

Gap 0

✓ You chose a correct answer. You were awarded **0.5** marks.

Gap 1

✓ Your answer is numerically correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

b)

Identify a matrix that applies a rotation of $\pi/4$ about the origin followed by a reflection through the line $y = mx$ with $m = 2$

Expected answer: $\begin{pmatrix} 0.141 & 0.990 \\ 0.990 & -0.141 \end{pmatrix}$

*Enter the entries of the matrix correct to 3 decimal places.

Score: 0.5/1 ✓

▼ Hide feedback

✓ One or more of the cells in your answer is incorrect, but you have been awarded marks for the rest. You were awarded **0.5** marks.

You scored **0.5** marks for this part.

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Module 3 Assessment

$$\begin{array}{r|l} \begin{array}{cccc} 1 & 2 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 7 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array} & \begin{array}{l} \xrightarrow{-R_2} \\ \xrightarrow{+5R_3} \\ \xrightarrow{-1} \\ \xrightarrow{-7R_3} \\ \xrightarrow{-1} \end{array} \end{array} \begin{array}{l} 100 \\ 010 \\ 001 \end{array} \begin{array}{l} -5 \ 1 \ 7 \\ 7 \ -1 \ -10 \\ -1 \ 0 \ 2 \end{array}$$

$$A = \begin{bmatrix} -5 & 9 \\ -7 & -6 \end{bmatrix} \quad B = \begin{bmatrix} -26 & -9 \\ -7 & -6 \end{bmatrix}$$

$$\begin{array}{r|l} \begin{array}{cc} 5 & 9 \\ 7 & -6 \end{array} & \begin{array}{l} \xrightarrow{\text{same}} \\ \xrightarrow{\times 5} \\ \xrightarrow{\times 7} \end{array} \end{array} \begin{array}{l} 100 \\ 010 \\ 001 \end{array} \begin{array}{l} -5 \ 1 \ 7 \\ 7 \ -1 \ -10 \\ -1 \ 0 \ 2 \end{array}$$

$$\begin{array}{r|l} \begin{array}{cccc} 1 & 0 & 0 & 1 \\ 0 & 1 & 6 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array} & \begin{array}{l} \xrightarrow{+6R_3} \\ \xrightarrow{+6R_4} \end{array} \end{array} \begin{array}{l} 100 \\ 010 \\ 001 \end{array} \begin{array}{l} 100 \\ 010 \\ 001 \end{array}$$

$$\begin{array}{r|l} \begin{array}{cccc} 1 & 0 & 0 & 1 \\ 0 & 1 & 6 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array} & \begin{array}{l} \xrightarrow{+6R_3} \\ \xrightarrow{+6R_4} \end{array} \end{array} \begin{array}{l} 100 \\ 010 \\ 001 \end{array} \begin{array}{l} 100 \\ 010 \\ 001 \end{array}$$

$$\begin{bmatrix} -26 & -9 \\ -7 & -6 \end{bmatrix} \begin{bmatrix} -\frac{6}{93} & -\frac{9}{93} \\ \frac{7}{93} & \frac{5}{93} \end{bmatrix} = \frac{1}{93} \begin{bmatrix} 156 & -63 \\ 42 & -30 \end{bmatrix}$$

$$T \begin{bmatrix} -2 \\ -8 \\ -6 \end{bmatrix} \quad T \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ -20 \end{bmatrix} \quad T \begin{bmatrix} -1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ -14 \end{bmatrix}$$

$$\frac{2m}{1+m^2} = -\frac{3}{5} \quad m = +3 \times$$

$$\begin{array}{l} 2x - y = -2 \therefore y = 2 \\ 4x + y = -8 \therefore x = -2 \\ x - y = -6 \end{array}$$

$$\begin{array}{l} -2 \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix} \\ 2 \begin{bmatrix} -1 \\ 0 \\ -2 \end{bmatrix} \end{array} = \begin{bmatrix} 0 \\ -8 \\ -2 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \\ -4 \end{bmatrix} = \begin{bmatrix} -2 \\ -8 \\ -6 \end{bmatrix}$$

$$\frac{6}{1+9} = \frac{-3}{5} \therefore m = -3$$

$$\begin{array}{l} -2 \begin{bmatrix} 10 \\ -20 \end{bmatrix} \\ 2 \begin{bmatrix} 3 \\ -14 \end{bmatrix} \end{array} = \begin{bmatrix} -20 \\ 40 \end{bmatrix}$$

$$\begin{bmatrix} -20 \\ 40 \end{bmatrix}$$

$$T = \begin{bmatrix} -\frac{4}{5} & -\frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{bmatrix}$$

Reflection

$$\begin{array}{l} 5(1-m^2) = -4(1+m^2) \\ 5-5m^2 = -4-4m^2 \\ 9 = m^2 \\ \therefore m = \pm 3 \end{array}$$

$$T = \frac{1}{1+m^2} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix} = \frac{-4}{5} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix} = \frac{-4}{5} \begin{bmatrix} -3 & 6 \\ 6 & -5 \end{bmatrix}$$

$\frac{\pi}{4}$ angle $m = 2$

$$\text{rotation} \begin{bmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \text{Rotation matrix}$$

$$\frac{1}{5} \times \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\rightarrow T = \begin{bmatrix} -\frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix} \times M = \begin{bmatrix} \frac{1}{5\sqrt{2}} & \frac{7}{5\sqrt{2}} \\ \frac{7}{5\sqrt{2}} & -\frac{1}{5\sqrt{2}} \end{bmatrix}$$

Reflexion matrix